

OFDG: a laptop-sized numerical preview

Initial evidence for choosing the next larger experiments

This brief compares accuracy, shock resolution, and positivity on small paper-inspired tests. It supports a supervisor discussion; it does not establish a universal method ranking.

What the completed runs suggest

Smooth accuracy: on the finest P2 grids, KXRCF gating keeps errors close to DG; ungated OFDG has larger errors.

Scalar shocks: filtering suppresses the large sampled DG excursions. The two OFDG variants are often nearly coincident.

Next decision: use Shu-Osher wave preservation and curved geometry to distinguish the methods more carefully before scaling up.

Experiment	Configuration	Question
Smooth advection	P1-P3; N=32/64/128/256; T=1.1	Accuracy across order
Piecewise advection	P2; N=128/256; T=1.1	Oscillations and smearing
Burgers	P2; smooth N=32/64/128, T=0.6; shock N=128/256, T=2.2	Nonlinear shock formation
Lax	P1/192, P2/128, P3/96; T=1.3	Equal unknown count
Shu-Osher	P2; N=200/400; T=1.8	Shock-wave interaction
Straight / curved transport	P2; 8 ² /16 ² /32 ² cells; T=1	Geometry and accuracy
2D Riemann	P2; 32 ² /64 ² cells; T=0.25	Wave structure and positivity

Methods and notation

DG is discontinuous Galerkin without the oscillation filter. **OFDG** is its oscillation-free variant; **OEDG** is the oscillation-eliminating comparator. **KXRCF gating** applies OFDG only in cells flagged by a troubled-cell indicator. Pk denotes polynomial degree k; N is the 1D cell count. Squared counts specify 2D grids.

All runs use RK4. CFL is 0.15 for scalar problems and 0.30 for Euler. OFDG filters after each full step; OEDG filters at RK stages. Every Euler method, including DG, uses the same positivity limiter. OEDG appears on selected finer affine shock cases only.

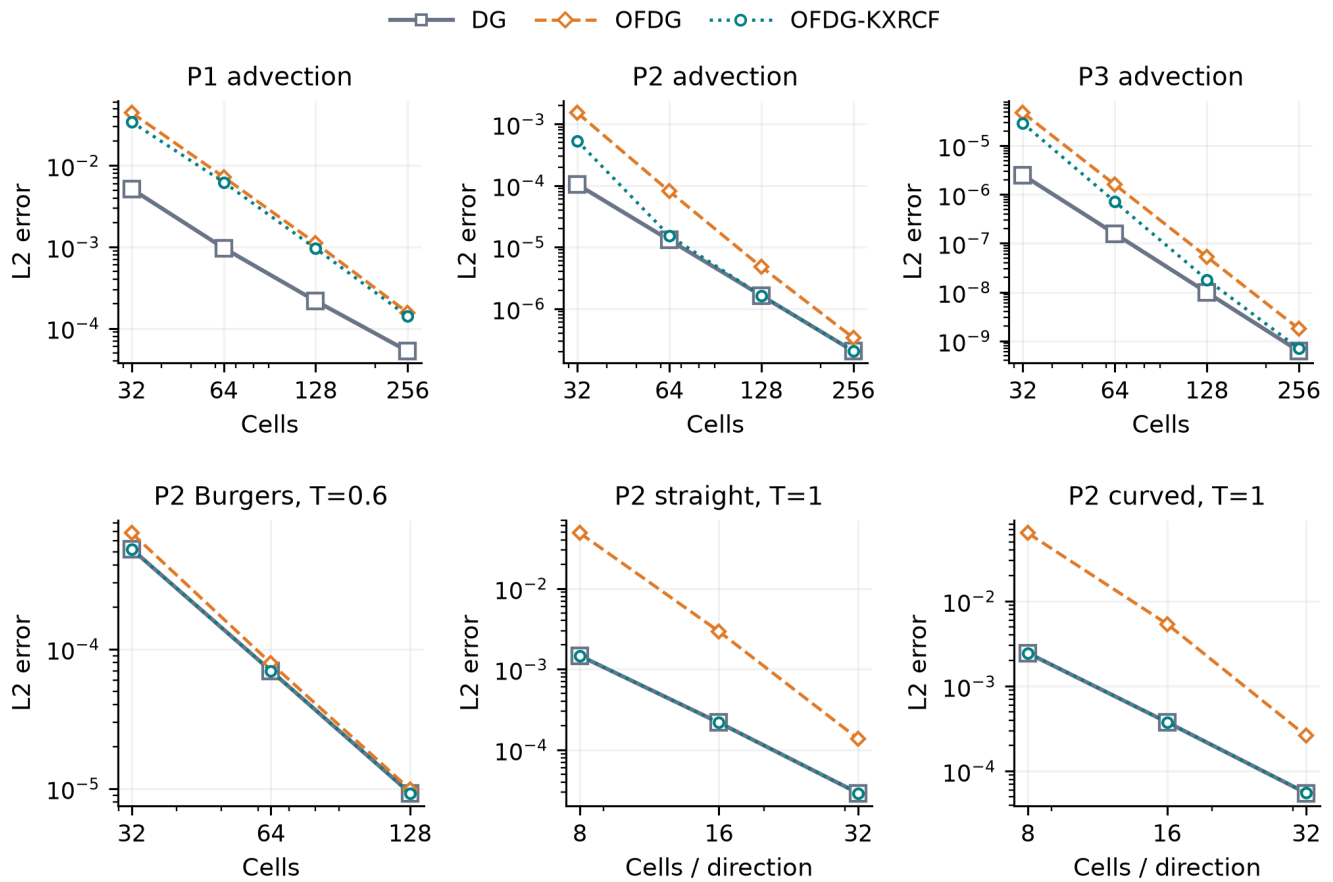
Compute used

103 solver runs and 4 reference solves completed in 7.35 minutes, with sampled peak process-tree memory 118 MiB. Runs were sequential and used at most two MPI ranks. No cases failed or were dropped.

Ceilings: 4 GB, five minutes per check, 60 minutes total; no new checks after 45 minutes. Memory peaks are sampled. Build and document preparation are excluded from solver time. Single-run timings do not establish performance differences.

Smooth accuracy and curved geometry

Refinement reduces every smooth error. Higher slopes in the filtered cases on these grids do not establish a higher asymptotic order.



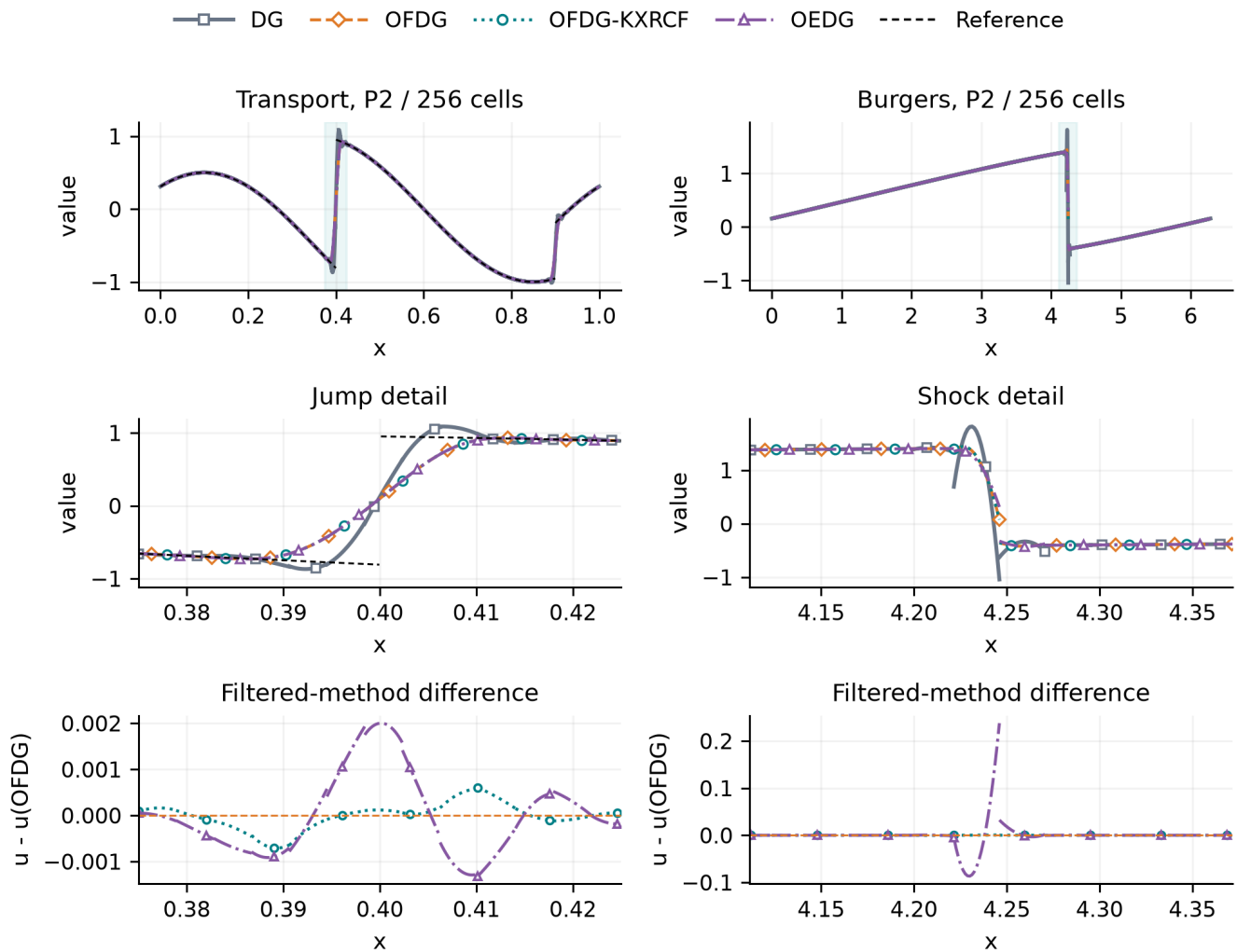
Method	P1 slope	P2 slope	P3 slope	Curved slope	P2 error / DG	Curved error / DG
DG	2.05	3.00	4.00	2.77	1.000	1.000
OFDG	2.83	3.83	4.90	4.37	1.648	4.715
OFDG-KXRCF	2.77	3.00	4.70	2.77	1.000	1.002

Slopes use the finest two grids. Nominal spatial orders are 2, 3, and 4 for P1, P2, and P3. Error ratios use $N=256$ (1D P2) and 32^2 (curved P2). Nested markers indicate nearly coincident curves: the table quantifies the hidden DG / gated-OFDG difference.

The straight and curved 2D panels share the unit-square transport problem: velocity $(0.7, 0.3)$, $\sin^2(\pi(x+y))$ data, and a static cubic deformation for the curved case. This is a 2021-paper-inspired adaptation.

Finest-P3 CFL halving changed L2 error by DG 0.0006%, OFDG 0.0029%, OFDG-KXRCF 0.0003%. No change exceeded 5%, so no extra series was needed. Maximum reported periodic scalar conservation drift: $4e-14$.

Scalar shocks: separate near-coincident curves



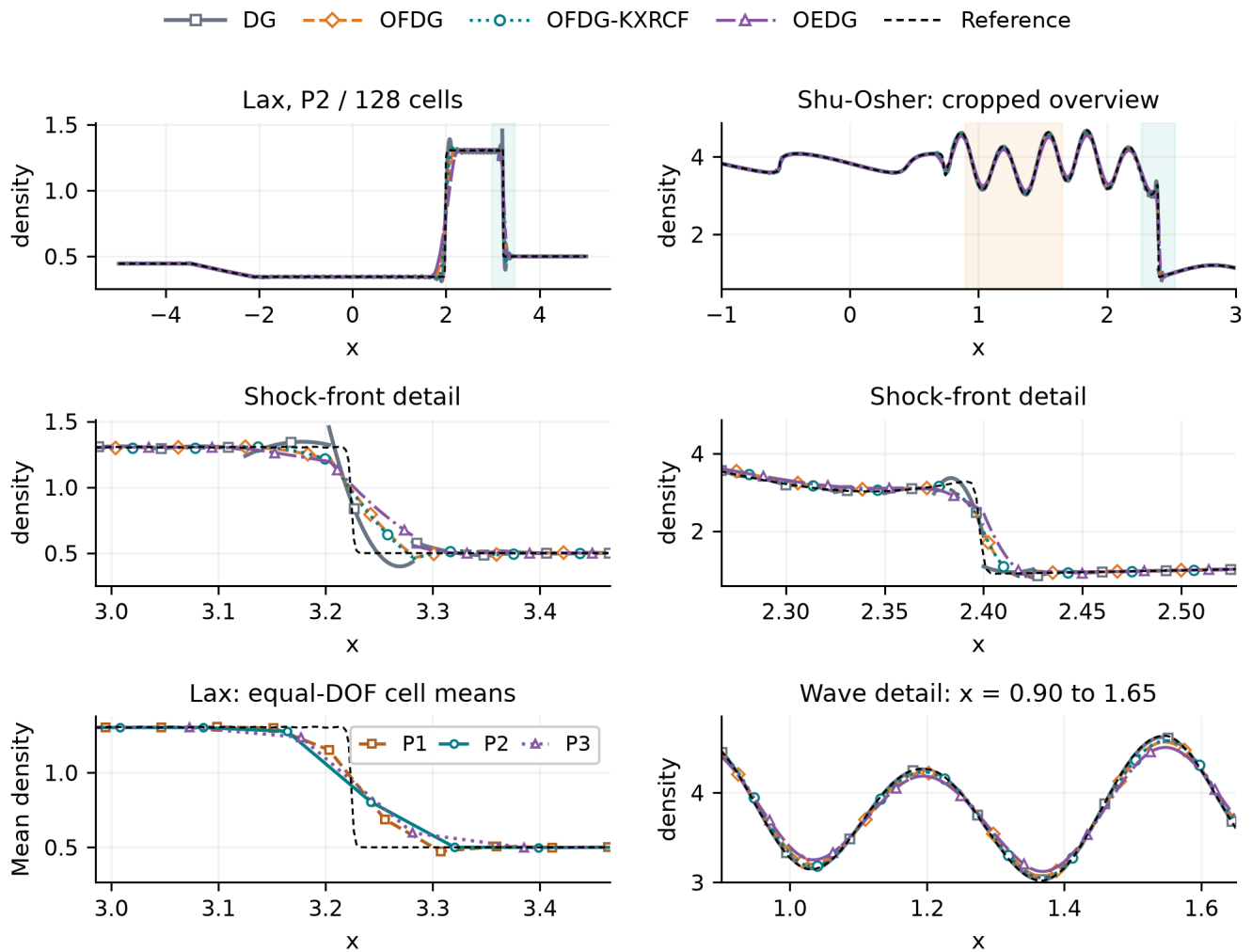
Overviews locate the shaded close-ups. The bottom row subtracts OFDG, so small differences between the filtered methods remain visible. The black reference is exact translated transport; Burgers has no reference curve.

Sampled excursion	Transport under	Transport over	Burgers under	Burgers over
DG	0.012	0.136	0.541	0.317
ODFG	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$
ODFG-KXRCF	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$
OEDG	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$

Excursions are beyond initial-data bounds: transport $[-1, 0.951057]$, Burgers $[-0.5, 1.5]$. Values below 10^{-6} are grouped for display; full precision is retained in analysis.json. The plots use 21 samples per cell, with curves broken at DG interfaces; sampled extrema are not certified continuous extrema.

In the displayed windows, $\max |\text{gated OFDG} - \text{ODFG}|$ is 0.00071 for transport and 0.00049 for Burgers. Near coincidence is a result here, not evidence that the curves were omitted.

Euler: shock fronts and wave detail



Blue bands mark shock details; the orange band marks the Shu-Osher wave detail. Shu-Osher uses P2 / 400 cells; its overview is cropped to [-1,3] from the [-5,5] domain. The equal-DOF Lax panel uses gated OFDG with P1/192, P2/128, P3/96: 384 scalar unknowns each. Its markers are cell means, not polynomial samples.

Wave-window peak-to-trough density range: DG 1.590, OFDG 1.495, OFDG-KXRCF 1.529, OEDG 1.383, WENO 1.615. This local range is a first look at amplitude; it does not measure phase accuracy.

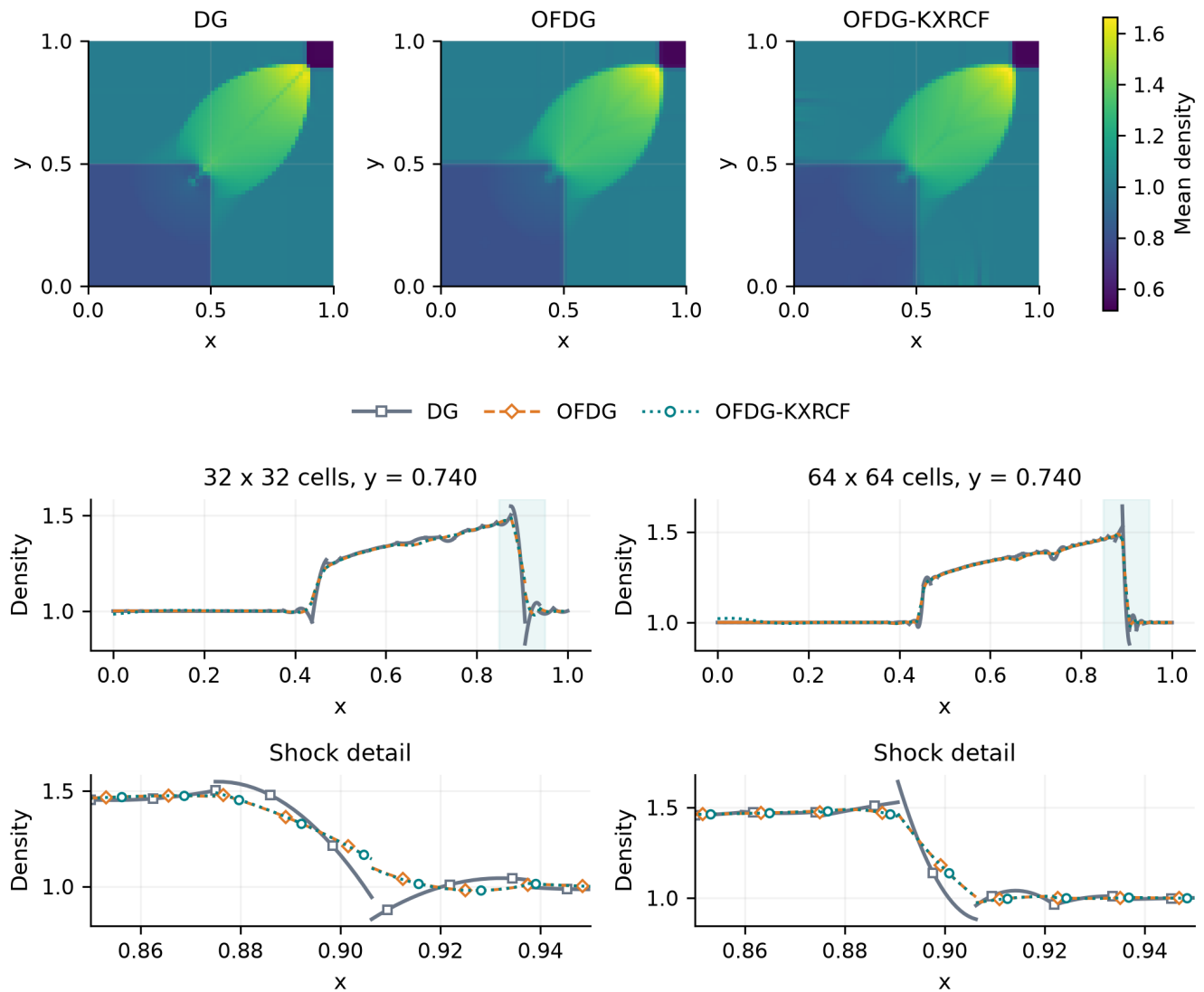
WENO reference pair	Mean absolute difference	Maximum difference
Lax 2048 vs 4096	0.00101	0.12
Shu-Osher 2048 vs 4096	0.00324	0.463

These are density differences after averaging the finer WENO reference onto the coarser grid. Large localized differences near steep fronts limit pointwise judgments even when mean differences are small. WENO5 / SSPRK3 is independent of DG, but is not exact.

Shu-Osher positivity cell-limit events: DG 6252, OFDG 3, OFDG-KXRCF 3, OEDG 2. These are cumulative interventions, not distinct cells. Fixed far-field boundaries permit changes in domain totals; those changes are not conservation-error estimates.

2D Riemann: compare the same physical cut

P2 Euler, $T=0.25$; four-quadrant data with outflow boundaries. Maps show 64^2 physical cell means on one shared scale. Its range spans all displayed means, so no values are clipped.



Both grids are evaluated at exactly $y=0.740$ by reconstructing each P2 polynomial from its tensor-node samples. Curves remain broken at cell interfaces. The two resolutions are separated to keep method comparisons readable.

The two OFDG traces nearly coincide. Max $|\text{gated OFDG} - \text{ODFG}|$ along this cut: 0.0173 on 32^2 and 0.0226 on 64^2 . The close-ups use polynomial traces; the maps above use cell means.

Method / cells	Min. density	Min. pressure	Limit events	Rejected steps
DG / 32^2	0.38	0.207	122	0
ODFG / 32^2	0.513	0.379	0	0
ODFG-KXRCF / 32^2	0.513	0.379	0	0
DG / 64^2	0.431	0.236	482	0
ODFG / 64^2	0.515	0.384	0	0
ODFG-KXRCF / 64^2	0.515	0.384	0	0

Minima are the solver-reported final values. Limit events count cumulative cell interventions. All three methods stayed admissible with the common positivity treatment; these coarse grids do not establish a resolution-independent ranking.

Decisions for the next compute allocation

Priority	Evidence from this preview	Next experiment / decision
1	Shu-Osher filters look close; amplitude alone cannot establish phase or wave accuracy.	Refine the wave region; measure amplitude and phase against a checked reference. Repeat timings only for this selected case.
2	Gating nearly matches DG smooth errors; curved errors are larger on the same grid counts.	Refine matched straight / curved problems, then a small curved Euler case. Separate geometry error from filtering.
3	2D OFDG variants differ little on the sampled cut, while DG uses more positivity interventions.	Refine the Riemann problem with boundary conditions and limiter settings matched across future comparisons.
Later	No result here requires a giant test to interpret.	Defer blast waves, 3D, and scaling studies until the scientific question and hardware are chosen.

What remains uncertain

This is a small configured comparison, not a full paper reproduction. Fluxes, initialization, filtering cadence, and positivity treatment affect the outcome. Only smooth P3 advection has a time-step sensitivity check. Shock widths and the wave-detail range need broader refinement. Native NURBS is outside the preview; curved results use supported polynomial geometry.

Open the evidence

Run: **preview**, 103 successful solver runs and four references.
Dataset: **measurements/study/preview/**
Instructions: **docs/supervisor-preview.md**
Configuration: **experiments/preview_manifest.json**

Paths are relative to `codex_dev`. The dataset retains exact commands, logs, physical samples, reference CSVs, convergence tables, and source/executable fingerprints. Existing thesis assets are unchanged. Project revision `f50a6dff8f98`; MFEM `a1ce49fb5742`.

Research basis

[1] J. Lu, Y. Liu, C.-W. Shu (2021). An Oscillation-Free Discontinuous Galerkin Method for Scalar Hyperbolic Conservation Laws. Scalar advection and Burgers examples. DOI: [10.1137/20M1354192](https://doi.org/10.1137/20M1354192)

[2] Y. Liu, J. Lu, C.-W. Shu (2022). An Essentially Oscillation-Free Discontinuous Galerkin Method for Hyperbolic Systems. Lax, Shu-Osher, and equal-DOF comparisons. DOI: [10.1137/21M140835X](https://doi.org/10.1137/21M140835X)

[3] M. Peng, Z. Sun, K. Wu (online 2024; issue 2025). OEDG: Oscillation-Eliminating Discontinuous Galerkin Method for Hyperbolic Conservation Laws. First four-quadrant Riemann setup. DOI: [10.1090/mcom/3998](https://doi.org/10.1090/mcom/3998)

Curved derivatives: does order matter?

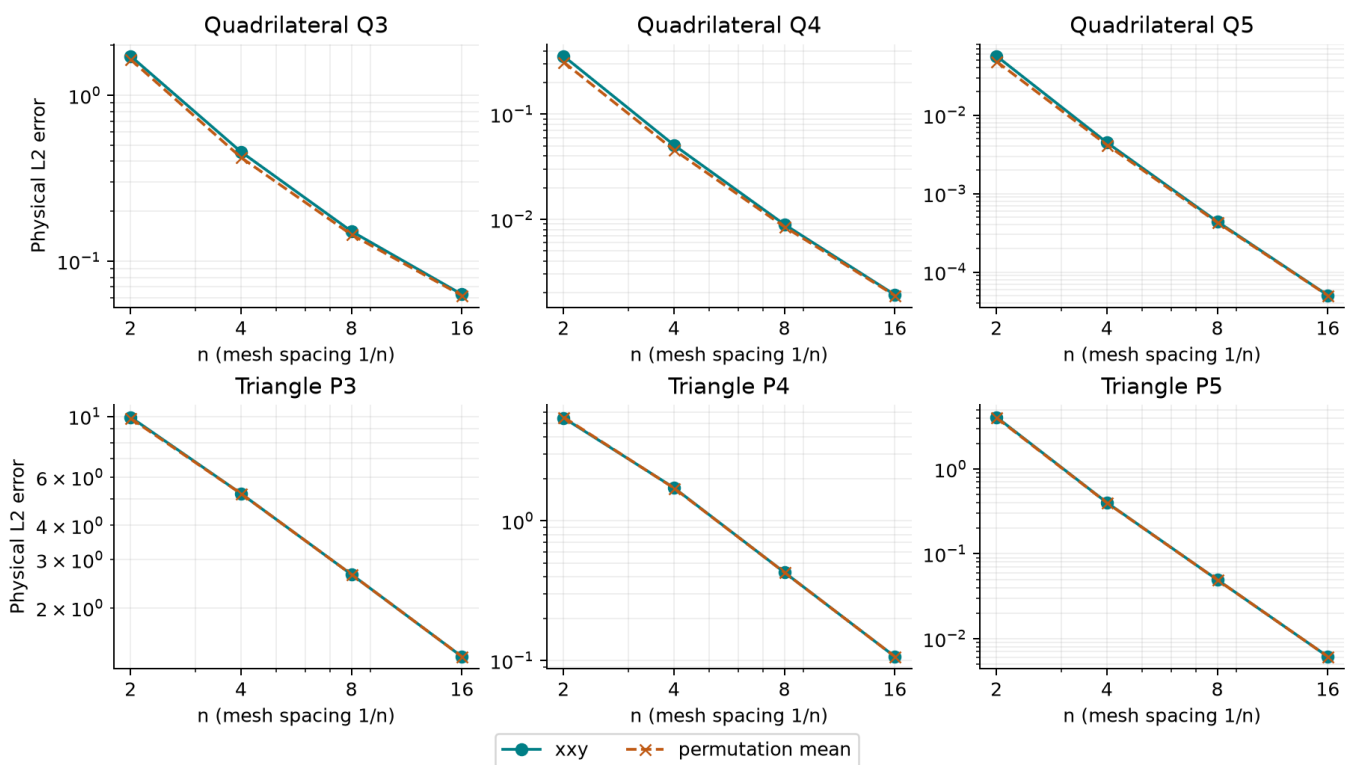
Isolated manufactured-derivative experiment; no PDE evolution and no change to the production OFDG algorithm. Exact mixed partial derivatives commute. This check measures the effect of inserting a physical L2 projection after each differentiation.

Construction and comparisons

Domain and geometry: start from (s,t) in $[0,1]^2$; set $b = a s(1-s)t(1-t)$, $x = s+b$ and $y = t+0.7b$. Use exact degree-4 polynomial geometry with $a = 0, 0.4, 1.2$ (affine, mild and stronger curvature). The domain boundary stays fixed; $\det J$ is at least $1-0.425a > 0$.

Spaces and data: triangle P3-P5 and quadrilateral Q3-Q5; $n = 2, 4, 8$, plus $n = 16$ at $a = 1.2$. Triangles have $2n^2$ elements, quadrilaterals n^2 ; these are not equal-DOF comparisons. Physically L2-project $\sin(2x+3y)$, $\exp(x+y/2)$, and x^2y+xy^2 into each space before differentiating.

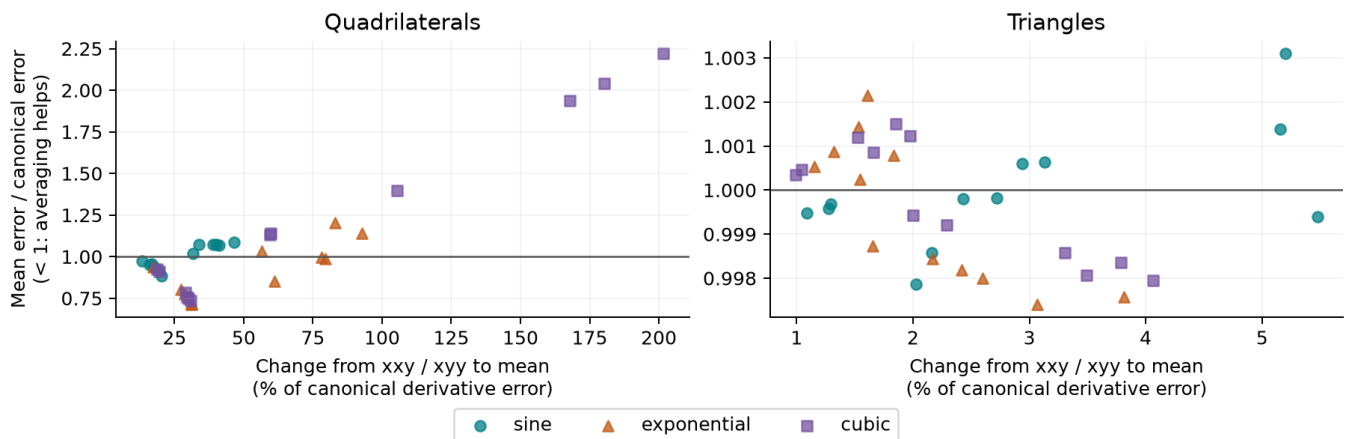
Sequences: words denote chronological operations: xy means x , then x , then y . Test xy/yx ; $xy/xyx/yxx$; and $xy/yxy/yxx$. The canonical sequence matches the production derivative graph. Average all distinct permutations as functions, then measure physical L2 errors against the analytic derivative. The average is a candidate approximation, not a reference solution.



Refinement: sine field, $a = 1.2$, xyx derivative. The exact derivative is $-12 \cos(2x+3y)$. Solid circles show the production order; dashed crosses show the permutation mean. Near-overlap is real; the next page plots changes directly. Both approaches improve under refinement in all six configurations.

Errors include differentiation of the initial solution projection error as well as the intermediate derivative projections. A separate direct projection of the analytic derivative supplies an approximation floor in the raw data. This is not a comparison with the exact higher derivative of the represented DG function.

Averaging: a change, not a guaranteed gain



Each point is one curved $n=8$ case: P/Q3-P/Q5, both nonzero curvature amplitudes, and xy or xyy . Horizontal position measures the L2 change to the derivative, divided by its canonical error. Vertical position measures the resulting error ratio. Points above 1 became less accurate after averaging. Axes use different scales: the triangle accuracy changes are much smaller.

Sine, $a=1.2$, $n=8$	Canonical error	Mean error	Error change
Q3 / xyy	0.1515	0.1446	-4.56%
Q4 / xyy	0.008933	0.008491	-4.95%
Q5 / xyy	0.0004397	0.0004283	-2.59%
P3 / xyy	2.649	2.649	-0.02%
P4 / xyy	0.4284	0.4278	-0.14%
P5 / xyy	0.04932	0.04922	-0.21%

Across 72 curved $n=8$ third-derivative cases, averaging lowers the error in 41. Mean/canonical error ratios range from 0.712 to 2.220 (median 0.999). The derivative change itself ranges from 1.0% to 201.6% of canonical error.

The largest worsening is mild-curvature Q3, cubic field, xyy : error 0.00109 to 0.00242 (exact derivative = 2). The relative errors are only 0.055% and 0.121%; the factor alone exaggerates its absolute significance.

Controls and interpretation

Affine canonical-to-mean differences are at most $5.6e-09$ in L2; the affine cubic derivative error is at most $3.3e-09$. Independent first-derivative weak residuals are below $3.4e-15$; constant-derivative coefficients below $8.5e-12$. Raising assembly quadrature by eight on four strong-curvature cases changes reported error norms by at most $2.4e-05$ relative.

Decision: retain the canonical implementation for now. Averaging removes the arbitrary ordering convention, but does not reliably reduce derivative error. These volume norms do not measure OFDG face-jump rates, damping, or final PDE accuracy. A targeted face-sensor and smooth-advection comparison is the next step before changing the method.

Resources and evidence: 64 sequential checks, one rank/thread; 4.63 minutes total, sampled peak 19.9 MiB. No new large PDE runs. Raw data, commands, hashes, quadrature controls and derived tables: [measurements/derivative-permutations/](#). Reproduction: [docs/derivative-permutations.md](#). This addendum is separate from the thesis.

How accurate are the derivatives?

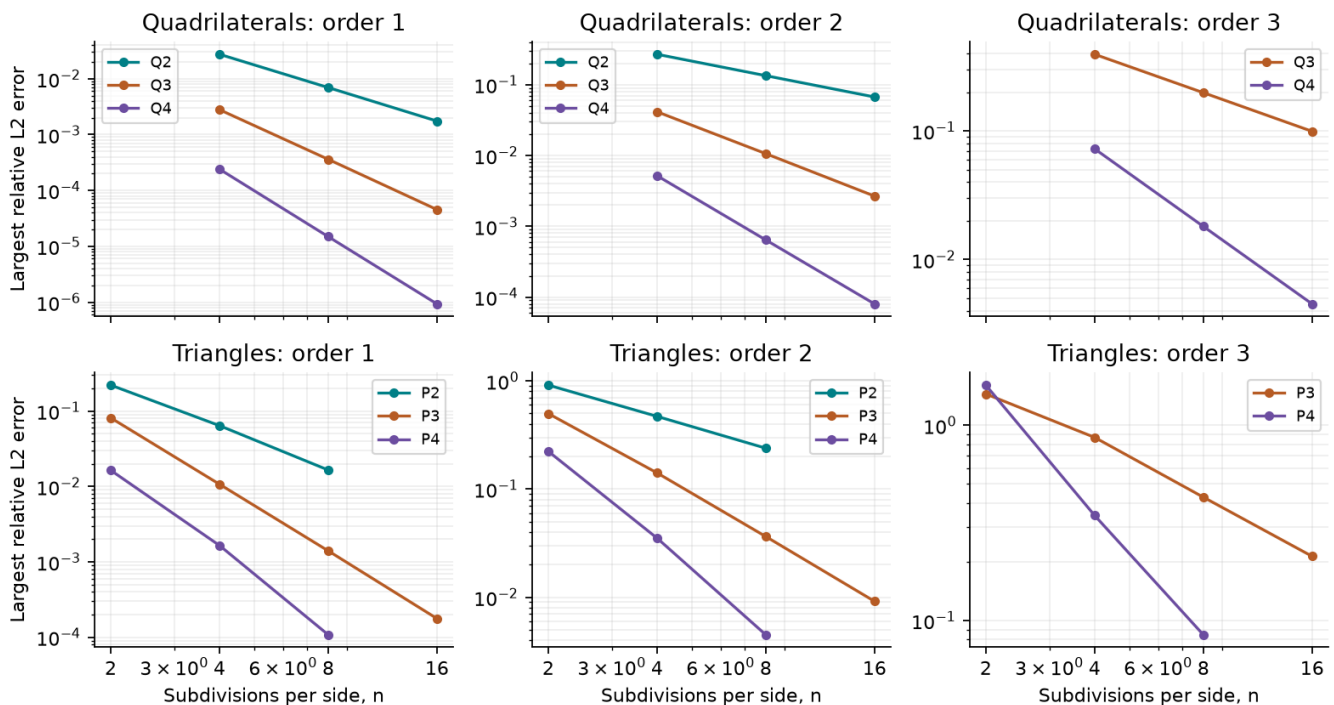
An isolated operator study compares the production projected-derivative matrices with analytic physical derivatives. No PDE is advanced and no production filtering code is changed. Higher derivative errors can be substantial even when first derivatives are accurate.

Two questions, tested separately

A. Complete approximation: physically project $\sin(2x+3y)$, $\sin(8x+6y)$, and $\exp(x+y/2)$, then differentiate recursively. Errors include the input projection and all intermediate derivative projections. Use the coupled curved-face map from page 7, with $a=0$, 0.4, 1.2.

B. Derivative projection alone: use an exactly representable mapped quadratic and an analytically invertible non-affine map (next page). This removes input approximation error and permits comparison with exact derivatives of the represented DG function itself.

Coverage: P/Q1-P/Q4 for A; P/Q2-P/Q4 for B. All canonical coordinate derivatives through $\min(p,3)$: $x, y, xx, xy, yy, xxx, xxy, xyy, yyy$. Gauss-Lobatto, Gauss-Legendre and positive Bernstein bases. Refinement uses $n=4,8,16$ for quads and $n=2,4,8$ for triangles, with selected $n=16$ triangle checks. Basis comparisons use $n=8$; the raw manifest identifies every case. Geometry degree 4 represents both maps exactly. Spaces at equal n are not equal-DOF comparisons.



Complete-approximation errors: $\sin(2x+3y)$, strong coupled curvature $a=1.2$, Gauss-Lobatto basis. Each curve shows the largest relative volume L2 error among derivatives of the indicated order. Different panel scales expose the refinement trend; high derivative orders remain much less accurate. These are canonical sequences, not permutation averages.

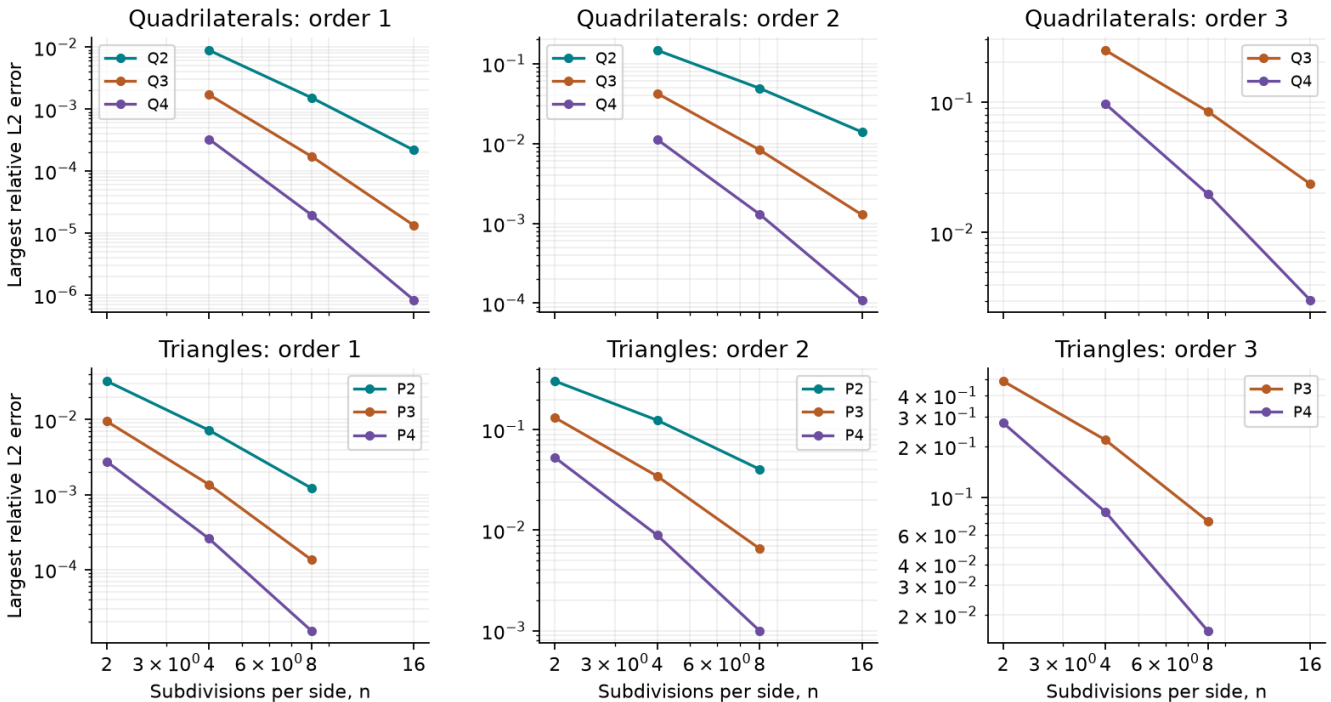
What “orders of magnitude off” means here

Absolute error $E = \|D^{\alpha} u_h - \text{partial}^{\alpha} u\|_{L_2}$; relative error $R = E / \|\text{partial}^{\alpha} u\|_{L_2}$. Thus $R=10^{-2}$ means a 1% norm error, two orders below the derivative norm. It is not a pointwise guarantee of two correct digits. Zero analytic derivatives have undefined relative error and are reported with absolute errors. No regularizing denominator hides those cases.

Separate projection error from input error

For the isolation test, take $x=s+a s(1-s)$, $y=t+0.7a t(1-t)$, and $u(x,y)=s(x)^2+s(x)t(y)+t(y)^2$. Test $a=0.3$ and 0.7 . On every element its pullback is quadratic, so P/Q2 and above represent it exactly. Its physical derivatives are generally nonpolynomial. The inverse map and its first three derivatives are analytic; see the reproducibility note.

The separable map stretches coordinate lines nonlinearly. Quad edges stay straight, while triangle diagonals can curve; it isolates non-affine parameterization rather than reproducing the curved quad faces of study A. Both maps are nonsingular throughout the domain.



Projection-only errors: strong separable map $a=0.7$. Initial physical projection error is at most $1.3e-14$ in L2 across the tested cases. All plotted errors can therefore be attributed to derivative representation, recursion and numerical roundoff, rather than an unresolved input field.

n=8, strong map	First derivative	Second derivative	Third derivative
Q2	3.08e-03 0.153% (x)	9.22e-01 4.91% (xx)	not used
Q3	3.47e-04 0.0173% (x)	1.56e-01 0.831% (xx)	5.25e+01 8.46% (xxx)
Q4	3.95e-05 0.00196% (x)	2.44e-02 0.13% (xx)	1.23e+01 1.98% (xxx)
P2	2.46e-03 0.122% (x)	7.58e-01 4.04% (xx)	not used
P3	2.73e-04 0.0136% (x)	1.23e-01 0.655% (xx)	4.48e+01 7.22% (xxx)
P4	3.08e-05 0.00153% (x)	1.87e-02 0.0997% (xx)	1.00e+01 1.61% (xxx)

Each cell gives absolute L2 error, then relative error; the displayed derivative has the largest relative error at that order. The worst absolute-error direction may differ. A direct physical projection of each exact derivative supplies the best-approximation floor. Higher recursive errors can exceed this floor because earlier projection errors are differentiated again. The raw results include both the floor and the excess over it.

Implications for OFDG and checks

How large are complete-approximation errors?

n=8; a=1.2	Order 1	Order 2	Order 3	Worst face error
Q2 / sine	1.48e-02 0.695%	3.78e-01 13.4%	not used	18.6%
Q3 / sine	7.67e-04 0.0361%	6.73e-02 1.06%	1.12e+00 19.8%	27.6%
Q4 / sine	3.17e-05 0.00149%	4.07e-03 0.0641%	1.03e-01 1.81%	3.2%
P2 / sine	2.34e-02 1.65%	6.73e-01 23.9%	not used	31%
P3 / sine	1.98e-03 0.14%	1.03e-01 3.64%	2.43e+00 42.8%	56.1%
P4 / sine	1.53e-04 0.0108%	1.27e-02 0.45%	4.80e-01 8.46%	13.8%

Volume cells show absolute L2 error / largest relative error over the canonical derivatives at each order. The last column is the largest relative one-sided face L2 error over all tested derivative orders, normalized by the analytic trace norm. It uses every element boundary, counting interior faces twice. It is a trace diagnostic, not the OFDG jump or damping rate.

Across these n=8 P/Q2-P/Q4 strong-curvature cases, the order-wise worst relative errors range from 2.6e-07 to 1.9 for the sine/exponential fields, and 0.00049 to 0.57 for the shorter wave. These are envelopes of a configured test set, not a statistical statement about typical meshes. Sampled maximum errors and nominal-h-scaled errors are also saved.

Basis, integration and analytic controls

Changing among Gauss-Lobatto, Gauss-Legendre and Bernstein bases changes the reported L2 error norms by at most 5e-08 absolute (8.4e-10 on nonconstant fields after scaling by the larger analytic derivative/input norm). This compares error norms, not pointwise equality of the derivative functions. Independent first-derivative weak residuals are at most 3.9e-15. Constant-field sampled derivative magnitudes reach 1.8e-06: repeated differentiation amplifies tiny initialization/assembly roundoff. Relative errors for these zero derivatives are undefined.

Raising assembly quadrature by eight changes error norms by at most 1.8e-10 on that scale; independently raising initialization/evaluation quadrature by eight changes them by 3.2e-10. Evaluation rules otherwise use production order +16. Analytic inverse-map derivatives are checked independently by high-precision finite differences. Sampled maxima are not certified continuous suprema.

Assessment: useful, with a resolution-dependent boundary

Retain this construction for OFDG, but do not describe its higher derivatives as uniformly accurate. Refinement and increased order improve the tested errors; changing coefficient basis does not cure them. First derivatives can be several orders below the true derivative norm while the highest derivative used by a low-order space remains at percent-to-tens-of-percent error. Strong distortion and shorter waves require particular care.

OFDG uses derivatives as roughness indicators, not as a replacement for the DG flux. That makes this a sensible construction to continue evaluating, but volume accuracy alone cannot establish acceptable damping or PDE accuracy. Face errors can be larger. Before claiming the extension is sufficiently accurate, compare face-jump rates and smooth-advection convergence against an independent derivative construction; shock robustness requires a separate check.

Resources: 135 completed checks, 0 failures; one rank/thread on an 8-logical-processor laptop. 14.69 minutes charged compute (including a conservative interruption allowance), peak sampled RSS 21.3 MiB. Raw data, exact commands, controls, source hashes and tables: [measurements/derivative-accuracy/](#). Instructions: [docs/derivative-accuracy.md](#). No new shock/PDE runs; no claim of stability or a universal error bound.